

APPENDIX A

DESCRIPTIVE STATISTICS AND DATA VISUALIZATION FOR MACHINE LEARNING



- Descriptive Statistics
 - Descriptive Statistics for Continuous Features
 - Descriptive Statistics for Categorical Features
 - Populations & Samples
- Data Visualization
 - Bar Plots
 - Histograms
 - Box Plots

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

1. Central Tendency

(1) Mean:

- The **arithmetic mean** (or **sample mean** or just **mean**) of a set of n values a_i ($i = 1, 2, \dots, n$) is calculated as:

$$\bar{a} = \frac{1}{n} \sum_{i=1}^n a_i$$

Example

ID	1	2	3	4	5	6	7	8
Height	150	163	145	140	157	151	140	149



Figure: The members of a school basketball squad. The dashed grey line shows the arithmetic mean of the players' heights.

$$\begin{aligned}\overline{\text{HEIGHT}} &= \frac{1}{8} \times (150 + 163 + 145 + 140 + 157 + 151 + 140 + 149) \\ &= 149.375\end{aligned}$$

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

1. Central Tendency

(1) Mean:

- The **arithmetic mean** is one measure of the **central tendency** of a **sample**
 - for our purposes, a sample is just a set of values for a feature in an ABT
- Any measure of **central tendency** is, however, just an approximation.

Example

- Suppose our basketball squad manage to sign a *ringer* measuring in at 229cm



- The arithmetic mean for the full group is 158.235cm and no longer represents the central tendency of the group.
- An unusually large or small value like this is referred to as an **outlier** - the arithmetic mean is very sensitive to outliers.

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

1. Central Tendency

(2) Median:

- The **median** of a set of values can be calculated by ordering the values from lowest to highest and selecting the middle value.
- If there is an **even number** of values in the sample then the median is obtained by calculating the arithmetic mean of the middle two values.

Example

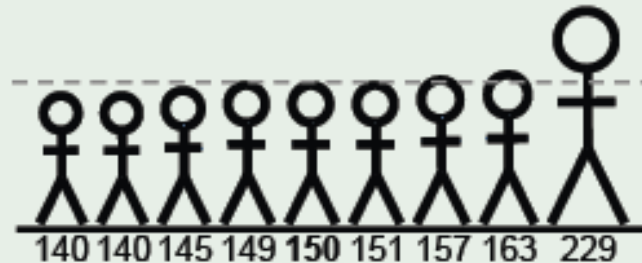


Figure: The members of the school basketball squad ordered by height, the dashed grey line shows the **median**.

ID	4	7	3	8	1	6	5	2	9
Height	140	140	145	149	<u>150</u>	151	157	163	229

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

2. Variation

(1) Range

- We also measure the variation in our data.
- In essence, most of statistics, and in turn analytics, is about describing and understanding variation.
- The simplest measure of variation is the **range**:

$$\text{range} = \max(a) - \min(a)$$

Example

What is the range of the heights of the two basketball squads?

ID	1	2	3	4	5	6	7	8
Height	150	163	145	140	157	151	140	149

ID	1	2	3	4	5	6	7	8
Height	192	102	145	165	126	154	123	188

Example

What is the range of the heights of the two basketball squads?

$$\text{range} = 163 - 140 = 23$$

$$\text{range} = 192 - 102 = 90$$

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

2. Variation

(2) Variance

- The **variance** of a sample measures the average difference between each value in a sample and the mean of that sample.
- The **variance** of the n values a_i ($i = 1, 2, \dots, n$) is calculated as:

$$\text{var}(a) = \frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n-1}$$

Example

What is the variance of the heights of the two basketball squads?

ID	1	2	3	4	5	6	7	8
Height	150	163	145	140	157	151	140	149

ID	1	2	3	4	5	6	7	8
Height	192	102	145	165	126	154	123	188

Example

$$\begin{aligned} \text{var}(\text{HEIGHT}) &= \frac{(150 - 149.375)^2 + (163 - 149.375)^2 + \dots + (149 - 149.375)^2}{8 - 1} \\ &= 63.125 \end{aligned}$$

$$\begin{aligned} \text{var}(\text{HEIGHT}) &= \frac{(192 - 149.375)^2 + (102 - 149.375)^2 + \dots + (188 - 149.375)^2}{8 - 1} \\ &= 1,011.41071 \end{aligned}$$

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

2. Variation

(3) Standard Deviation

- The **standard deviation** of a sample is calculated by taking the square root of the **variance** of the sample:

$$sd(a) = \sqrt{\text{var}(a)} = \sqrt{\frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n-1}}$$

Example

What is the standard deviation of the heights of the two basketball squads?

ID	1	2	3	4	5	6	7	8
Height	150	163	145	140	157	151	140	149

ID	1	2	3	4	5	6	7	8
Height	192	102	145	165	126	154	123	188

Example

$$\begin{aligned}sd(\text{HEIGHT}) &= \sqrt{63.125} \\ &= 7.9451 \dots\end{aligned}$$

$$\begin{aligned}sd(\text{HEIGHT}) &= \sqrt{1,011.41071} \\ &= 31.8026 \dots\end{aligned}$$

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

2. Variation

(4) Percentiles

- **Percentiles** are another useful measure of the variation of the values for a feature
- A **proportion** of $i/100$ of the values in a sample take values equal to or lower than the i^{th} percentile of that sample.

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

2. Variation

(4) Percentiles

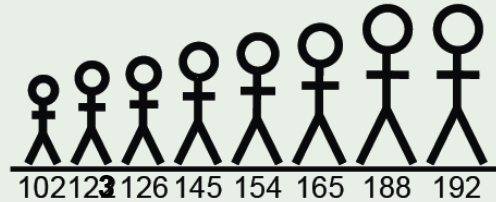
- To calculate the i^{th} percentile of the n values a_i ($i = 1, 2, \dots, n$):
 - First order the values in **ascending order** and then multiply n by $i/100$ to determine the **index**.
 - If the **index** is a whole number, we take the value at that position in the ordered list of values as the i^{th} percentile.
 - If **index** is not a whole number, then we **interpolate** the value for the i^{th} percentile as:

$$i^{\text{th}} \text{ percentile} = (1 - \text{index_}f) \times a_{\text{index_}w} + \text{index_}f \times a_{\text{index_}w+1}$$

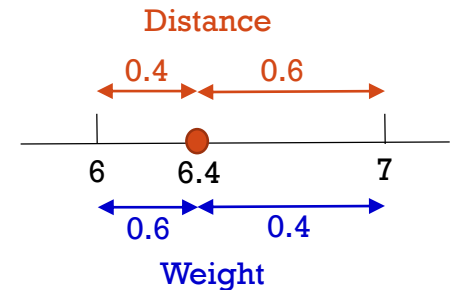
where $\text{index_}w$ is the **w**hole part of **index**, $\text{index_}f$ is the **f**ractional part of **index** and $a_{\text{index_}w}$ is the value in the ordered list at position $\text{index_}w$.

Example

ID	2	7	5	3	6	4	8	1
Height	102	123	126	145	154	165	188	192



- What is the 25th percentile of the heights of the basketball squad?
- What is the 80th percentile of the heights of the basketball squad?



- To calculate the 25th percentile, we first calculate *index* as $0.25 \times 8 = 2$. So, the 25th percentile is the second value in the ordered list which is 123.
- To calculate the 80th percentile, we first calculate *index* as $0.80 \times 8 = 6.4$. Because *index* is not a whole number, we set *index_w* to the whole part of *index*, 6, and *index_f* to the fractional part, 0.4. Then we can calculate the 80th percentile as:

$$(1-0.4) \times 165 + 0.4 \times 188 = 174.2$$

DESCRIPTIVE STATISTICS FOR CONTINUOUS FEATURES

2. Variation

(5) Inter-Quartile Range

- The **inter-quartile range** is calculated as the difference between the 25th percentile and the 75th percentile.
 - 25th percentile is also known the **lower quartile** (or 1st quartile)
 - 75th percentile is also known **upper quartile** (or 3rd quartile) hence the name inter-quartile range

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DESCRIPTIVE STATISTICS FOR CATEGORICAL FEATURES

Frequency counts and Proportions

- For categorical features we are interested primarily in **frequency counts** and **proportions**.
 - The **frequency count** of each level of a categorical feature is calculated by counting the number of times that level appears in the sample.
 - The **proportion** for each level is calculated by dividing the frequency count for that level by the total sample size.
 - Frequencies and proportions are typically presented in a **frequency table**.

DESCRIPTIVE STATISTICS FOR CATEGORICAL FEATURES

Mode

- The **mode** is a measure of the central tendency of a categorical feature and is simply the most frequent level.
- We often also calculate a **second mode** which is just the second most common level of a feature.

DESCRIPTIVE STATISTICS FOR CATEGORICAL FEATURES

- **TableA.1:** A dataset showing the positions and weekly training expenses of a school basketball squad.

ID	Position	Training Expenses	ID	Position	Training Expenses
1	center	56.75	11	center	550.00
2	guard	1,800.11	12	center	223.89
3	guard	1,341.03	13	center	103.23
4	forward	749.50	14	forward	758.22
5	guard	1,150.00	15	forward	430.79
6	forward	928.30	16	forward	675.11
7	center	250.90	17	guard	1,657.20
8	guard	806.15	18	guard	1,405.18
9	guard	1,209.02	19	guard	760.51
10	forward	405.72	20	forward	985.41

DESCRIPTIVE STATISTICS FOR CATEGORICAL FEATURES

- **Table A.2:** A frequency table for the POSITION feature from the professional basketball squad dataset

Level	Count	Proportion
guard	8	40%
forward	7	35%
center	5	25%

The **mode** of the POSITION is *guard*, the **second mode** is *forward*.

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POPULATIONS AND SAMPLES

- In statistics it is very important to understand the **difference between** a **population** and a **sample**.
- The term **population** is used in statistics to represent **all** possible measurements or outcomes that are of interest to us in a particular study or piece of analysis.
- The term **sample** refers to the **subset** of the population that is selected for analysis.
- The **margin of error** reported in poll results takes into account the fact that the result is based on a sample from a much larger population.

POPULATIONS AND SAMPLES

- **Table A.3:** A number of poll results from the run up to the 2012 US Presidential election.

Poll	Obama	Romney	Other	Date	Margin of Error	Sample Size
Pew Research	50	47	3	04-Nov	±2.2	2,709
Gallup	49	50	1	04-Nov	±2.0	2,700
ABC News/Wash Pos	50	47	3	04-Nov	±2.5	2,345
CNN/Opinion Research	49	49	2	04-Nov	±3.5	963
Pew Research	50	47	3	03-Nov	±2.2	2,709
ABC News/Wash Post	49	48	3	03-Nov	±2.5	2,069
ABC News/Wash Post	49	49	2	30-Oct	±3.0	1,288

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DATA VISUALIZATION

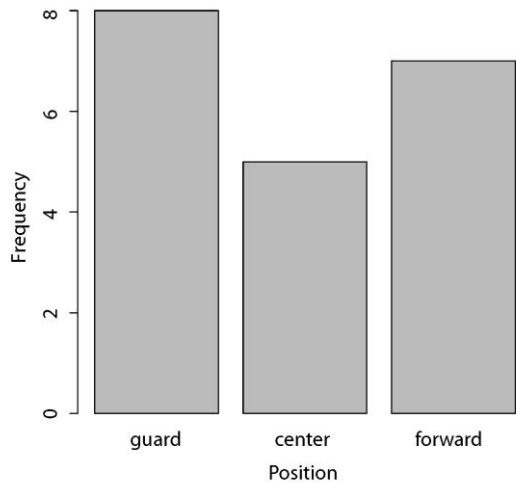
- **Table A.1:** A dataset showing the positions and weekly training expenses of a school basketball squad.

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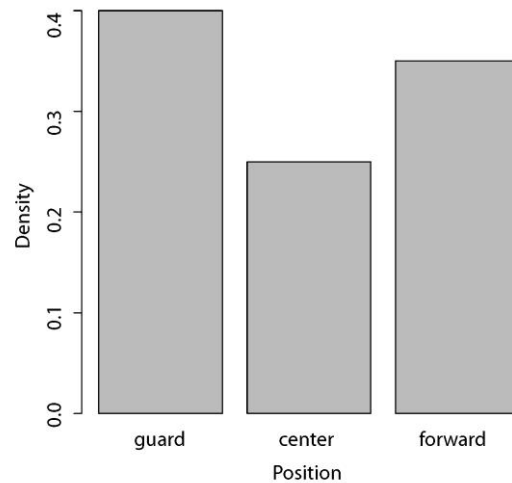
BAR PLOTS

- **Bar plots** are great for **categorical** features, but don't work for continuous features

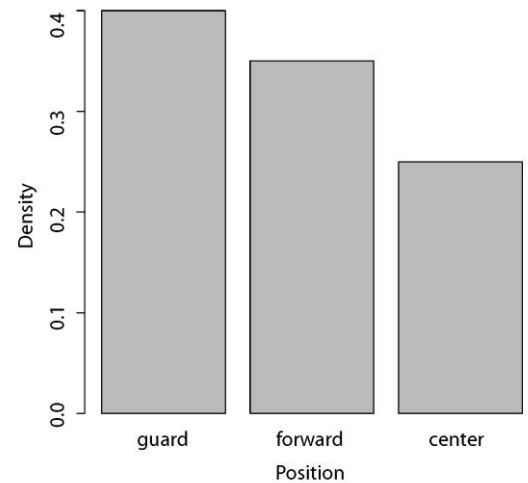
Figure A.5 Example bar plots for the POSITION feature (a) frequency bar plot, (b) density bar plot, and (c) order density bar plot.



(a) Frequency



(b) Proportion



(c) Ordered

HISTOGRAMS

- By dividing the range of a variable into intervals, or bins, we can generate **histograms**

(a) 200 unit intervals

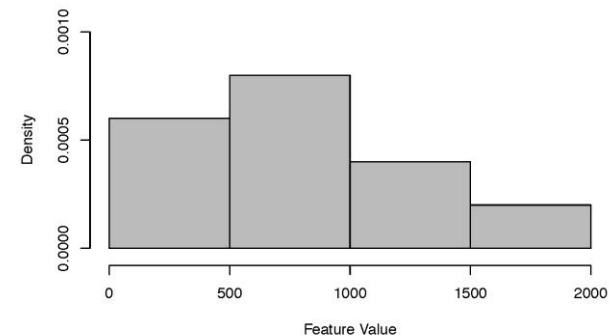
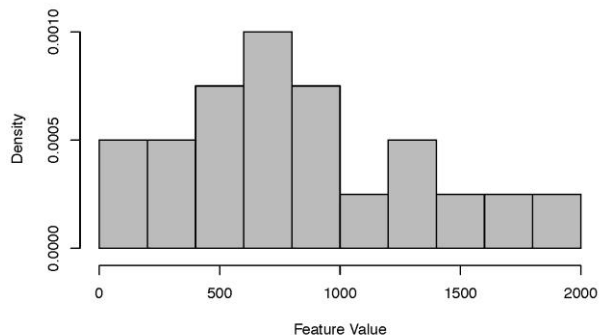
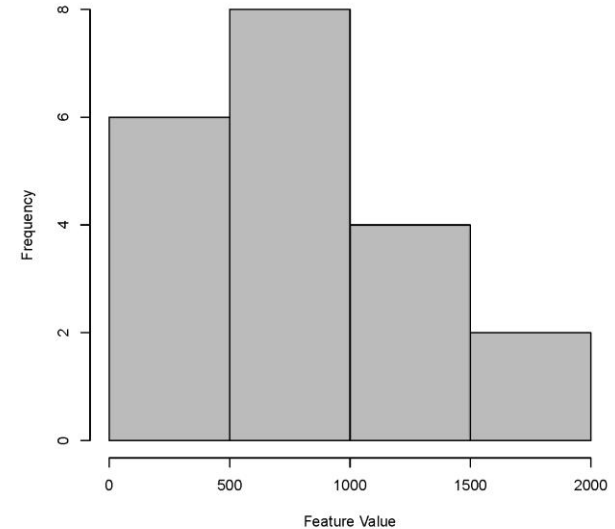
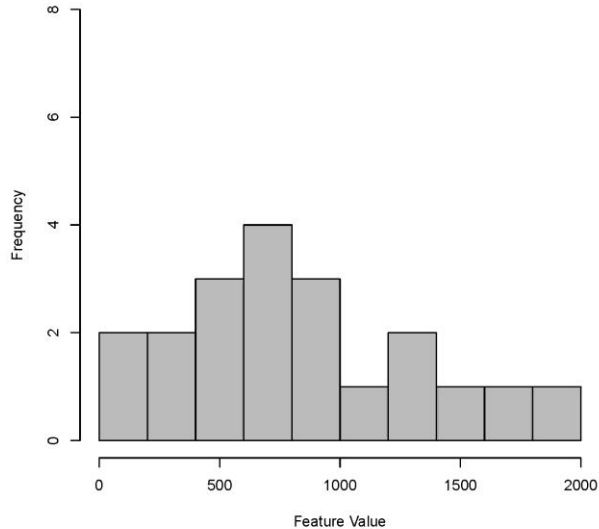
Interval	Count	Density	Prob
[0, 200)	2	0.0005	0.1
[200, 400)	2	0.0005	0.1
[400, 600)	3	0.00075	0.15
[600, 800)	4	0.001	0.2
[800, 1000)	3	0.00075	0.15
[1000, 1200)	1	0.00025	0.05
[1200, 1400)	2	0.0005	0.1
[1400, 1600)	1	0.00025	0.05
[1600, 1800)	1	0.00025	0.05
[1800, 2000)	1	0.00025	0.05

(b) 500 unit intervals

Interval	Count	Density	Prob
[0, 500)	6	0.0006	0.3
[500, 1000)	8	0.0008	0.4
[1000, 1500)	4	0.0004	0.2
[1500, 2000)	2	0.0002	0.1

HISTOGRAMS

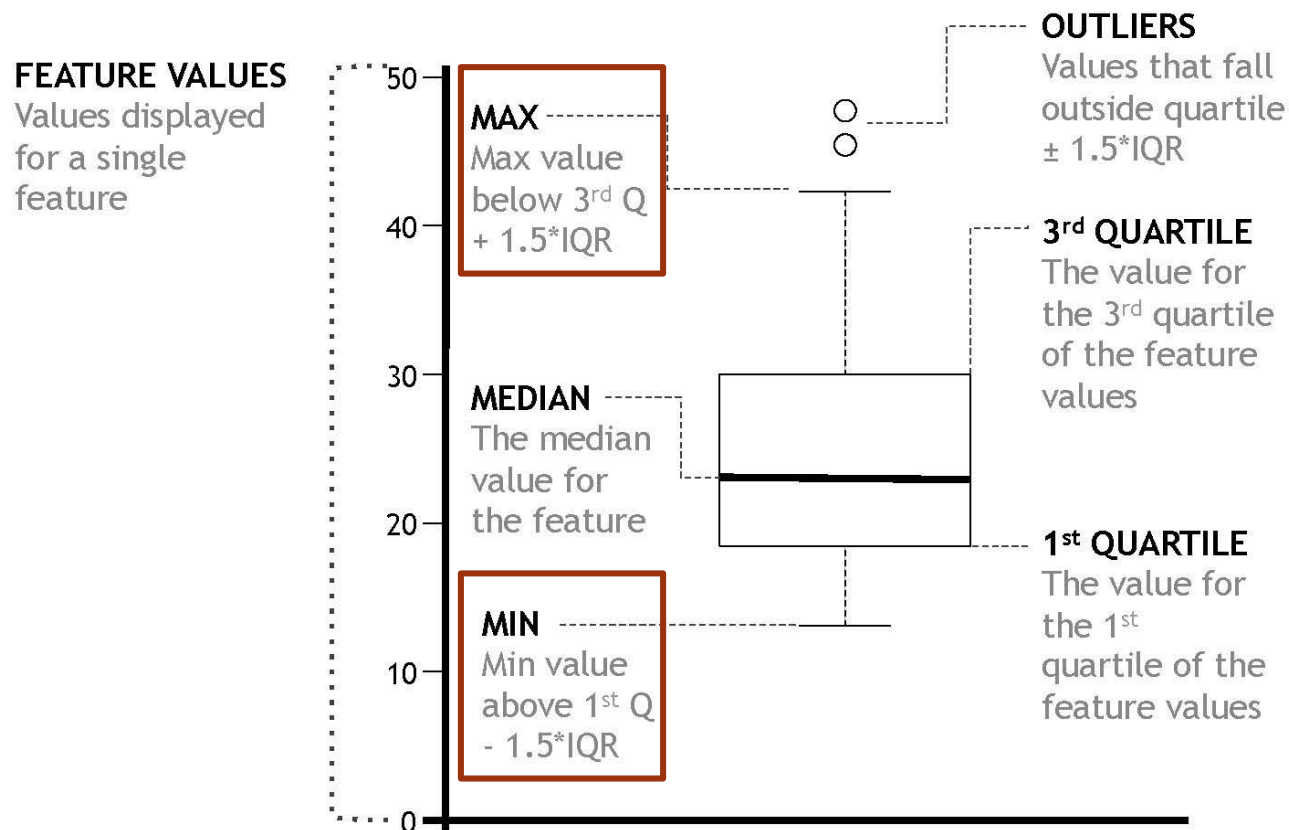
- **Figure A.7:** Frequency and density histograms for the continuous Training Expenses feature



BOX PLOTS

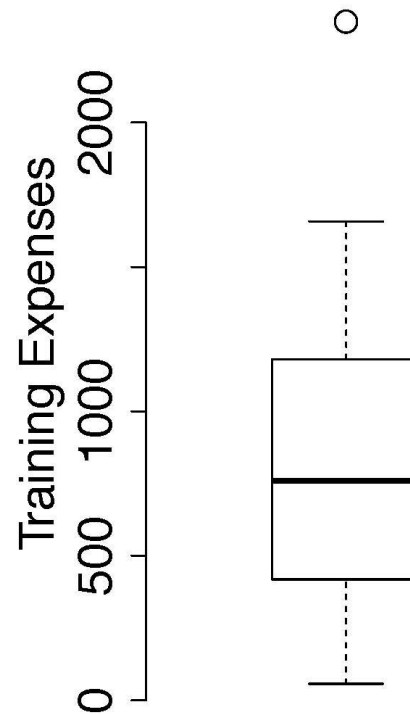
- **Box plots** are another useful way of visualising **continuous** variables

Figure A.8 (a) The structure of a box plot



BOX PLOTS

Figure A.8(b): A box plot for the TRAINING EXPENSES feature from the basketball squad dataset

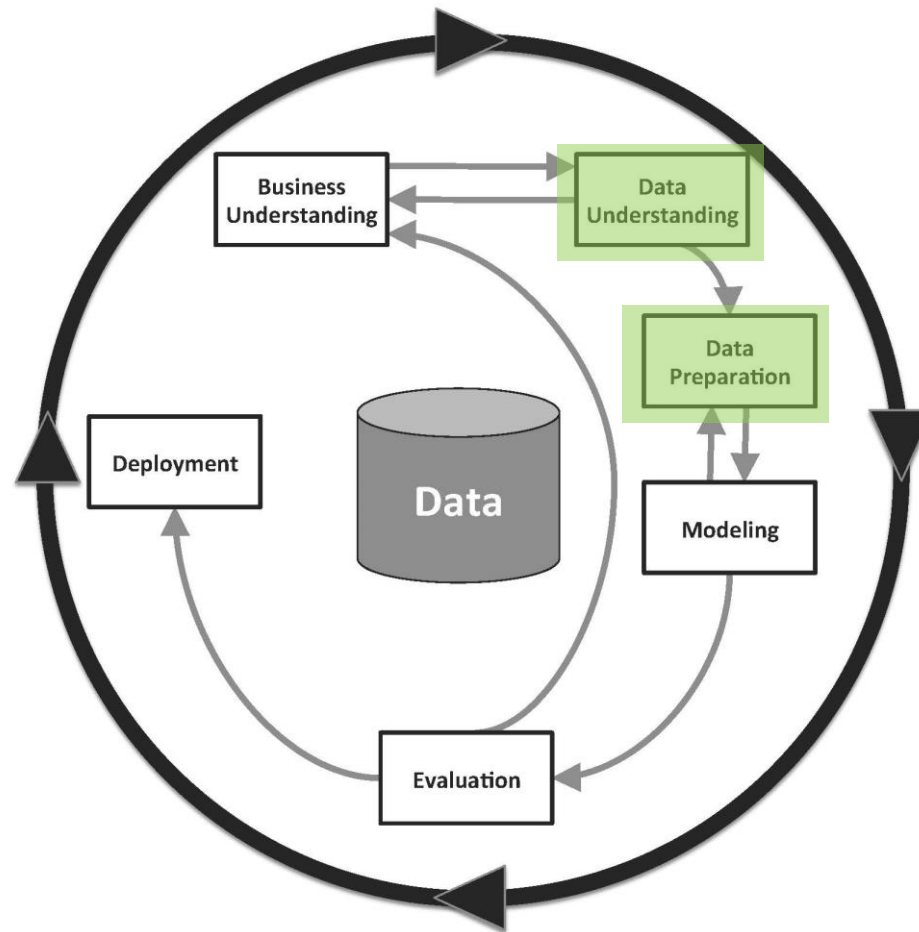


3. DATA EXPLORATION (1)

SECTION 3.1~3.4



- **Data exploration** is a key part of both the **Data Understanding** and **Data Preparation** phases of CRISP-DM.



CONTENTS

- **The Data Quality Report**
- **Getting To Know The Data**
- **Identifying Data Quality Issues**
- **Handling Data Quality Issues**
- **Summary**

- **A data quality report** includes tabular reports that describe the characteristics of each feature in an ABT using **standard statistical measures** of **central tendency** and **variation**.
- The tabular reports are accompanied by **data visualizations**:
 - A **histogram** for each **continuous** feature in an ABT.
 - A **bar plot** for each **categorical** feature in an ABT.

Table 3.1 : The structures of the tables included in a **data quality report** to describe (a) **continuous features** and (b) **categorical features**.

cardinality measures **the number of distinct values** present in the ABT for a feature

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min.	1 st Qrt.	Mean	Median	3 rd Qrt.	Max.	Std. Dev.
—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—

(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—

the mode and 2nd mode are **the two most frequent levels** for the feature

CASE STUDY: MOTOR INSURANCE FRAUD

- The following slides show a portion of the ABT that has been developed for the motor insurance claims fraud detection.

- **Table 3.2:** Portions of the **ABT** for the motor insurance claims fraud detection problem.

ID	TYPE	INC.	MARITAL STATUS	NUM CLMNTS.	INJURY TYPE	HOSPITAL STAY	CLAIM AMNT.	TOTAL CLAIMED	NUM CLAIMS	NUM SOFT TISS.	% SOFT TISS.	CLAIM AMT RCVD.	FRAUD FLAG
1	CI	0	Married	2	Soft Tissue	No	1,625	3250	2	2	1.0	0	1
2	CI	0		2	Back	Yes	15,028	60,112	1		0	15,028	0
3	CI	54,613		1	Broken Limb	No	-99,999	0	0	0	0	572	0
4	CI	0		4	Broken Limb	Yes	5,097	11,661	1	1	1.0	7,864	0
5	CI	0	Single	4	Soft Tissue	No	8869	0	0	0	0	0	1
6	CI	0		1	Broken Limb	Yes	17,480	0	0	0	0	17,480	0
7	CI	52,567		3	Broken Limb	No	3,017	18,102	2	1	0.5	0	1
8	CI	0		2	Back	Yes	7463	0	0	0	0	7,463	0
9	CI	0	Married	1	Soft Tissue	No	2,067	0	0	0	0	2,067	0
10	CI	42,300		4	Back	No	2,260	0	0	0	0	2,260	0
⋮													
300	CI	0	Married	2	Broken Limb	No	2,244	0	0	0	0	2,244	0
301	CI	0		1	Broken Limb	No	1,627	92,283	3	0	0	1,627	0
302	CI	0		3	Serious	Yes	270,200	0	0	0	0	270,200	0
303	CI	0		1	Soft Tissue	No	7,668	92,806	3	0	0	7,668	0
304	CI	46,365		1	Back	No	3,217	0	0		0	1,653	0
⋮													
458	CI	48,176	Married	3	Soft Tissue	Yes	4,653	8,203	1	0	0	4,653	0
459	CI	0	Divorced	1	Soft Tissue	Yes	881	51,245	3	0	0	0	1
460	CI	0		3	Back	No	8,688	729,792	56	5	0.08	8,688	0
461	CI	47,371		1	Broken Limb	Yes	3,194	11,668	1	0	0	3,194	0
462	CI	0		1	Soft Tissue	No	6,821	0	0	0	0	0	1
⋮													
491	CI	40,204	Single	1	Back	No	75,748	11,116	1	0	0	0	1
492	CI	0	Married	1	Broken Limb	No	6,172	6,041	1		0	6,172	0
493	CI	0		1	Soft Tissue	Yes	2,569	20,055	1	0	0	2,569	0
494	CI	31,951		1	Broken Limb	No	5,227	22,095	1	0	0	5,227	0
495	CI	0		2	Back	No	3,813	9,882	3	0	0	0	1
496	CI	0	Married	1	Soft Tissue	No	2,118	0	0	0	0	0	1
497	CI	29,280		4	Broken Limb	Yes	3,199	0	0	0	0	0	1
498	CI	0	Married	1	Broken Limb	Yes	32,469	0	0	0	0	16,763	0
499	CI	46,683		1	Broken Limb	No	179,448	0	0		0	179,448	0
500	CI	0		1	Broken Limb	No	8,259	0	0	0	0	0	1

- **Table 3.3:** A [data quality report](#) for the motor insurance claims fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

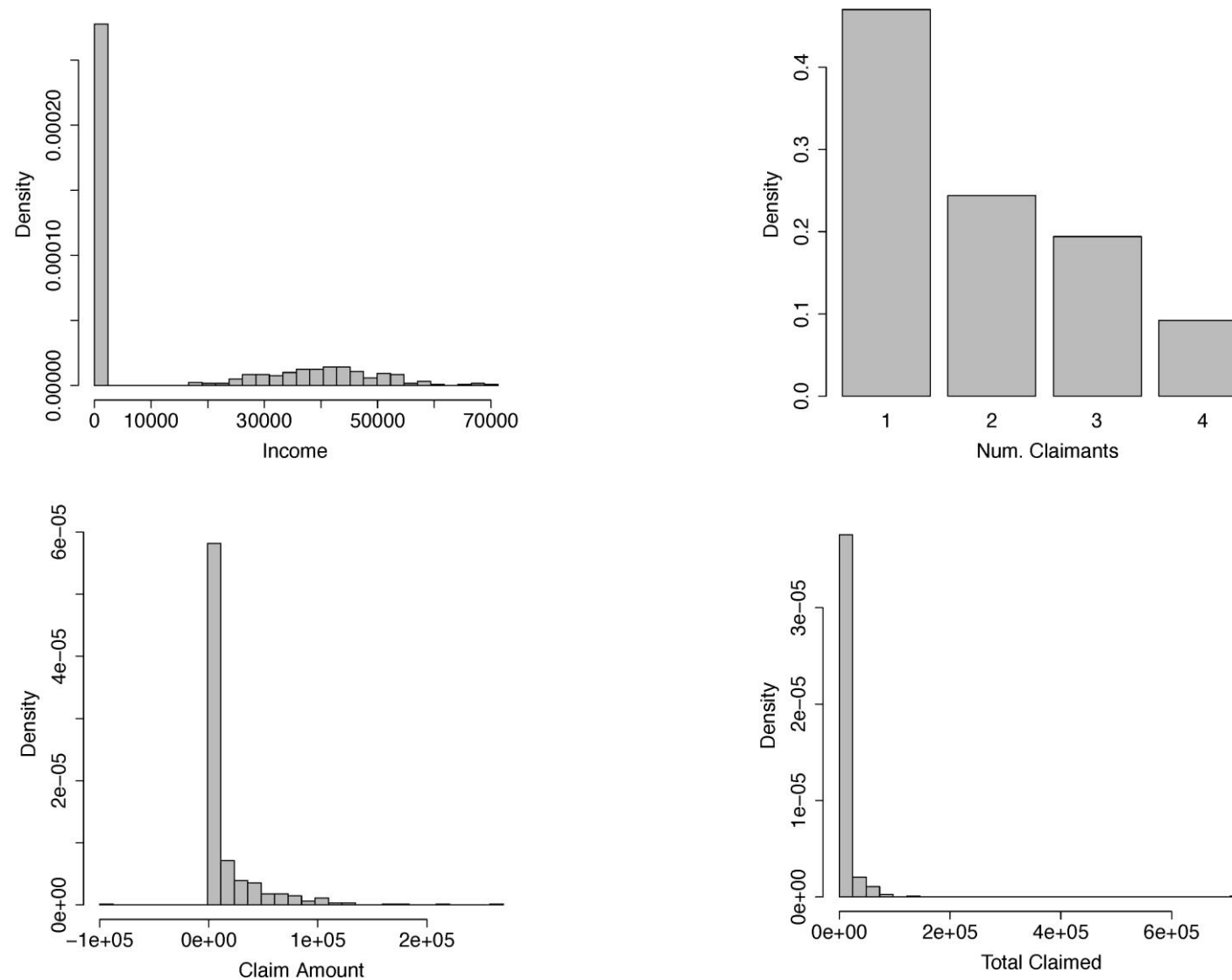


Figure 3.1(a~d): Visualizations of the continuous and categorical features in the motor insurance claims fraud detection ABT in Table 3.2

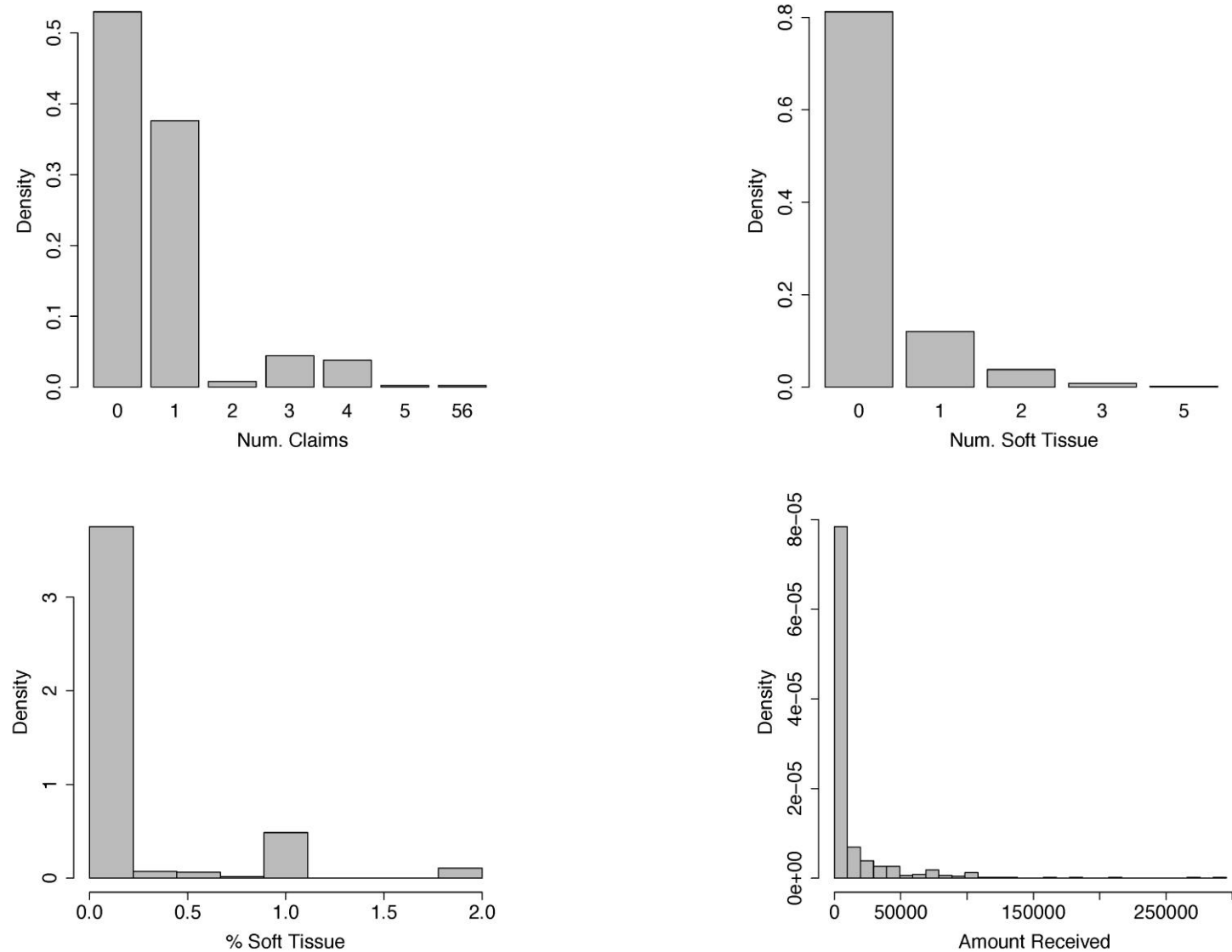


Figure 3.1(e~h): Visualizations of the continuous and categorical features in the motor insurance claims fraud detection ABT in Table 3.2

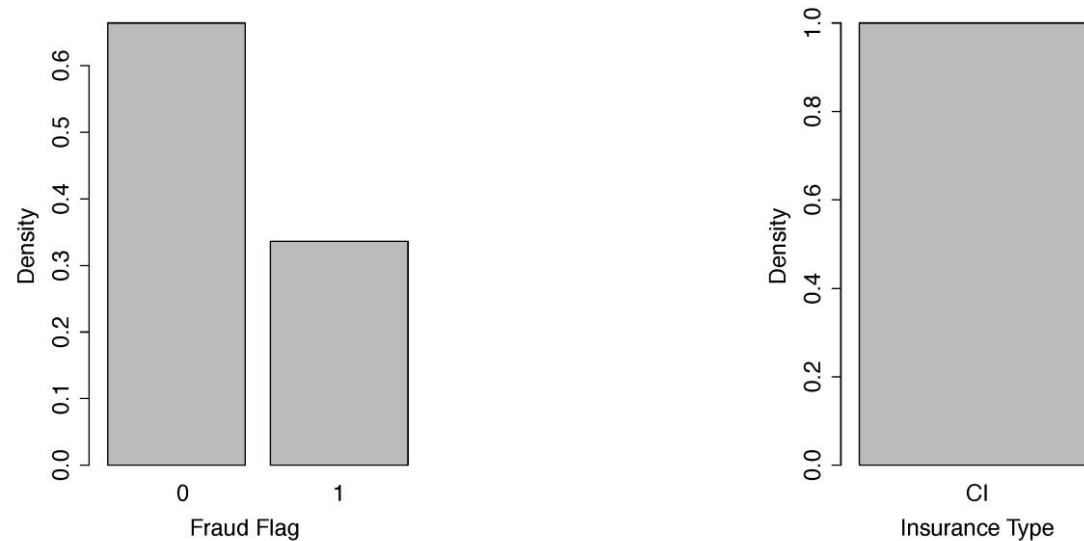


Figure 3.1(i, j): Visualizations of the continuous and categorical features in the motor insurance claims fraud detection ABT in Table 3.2

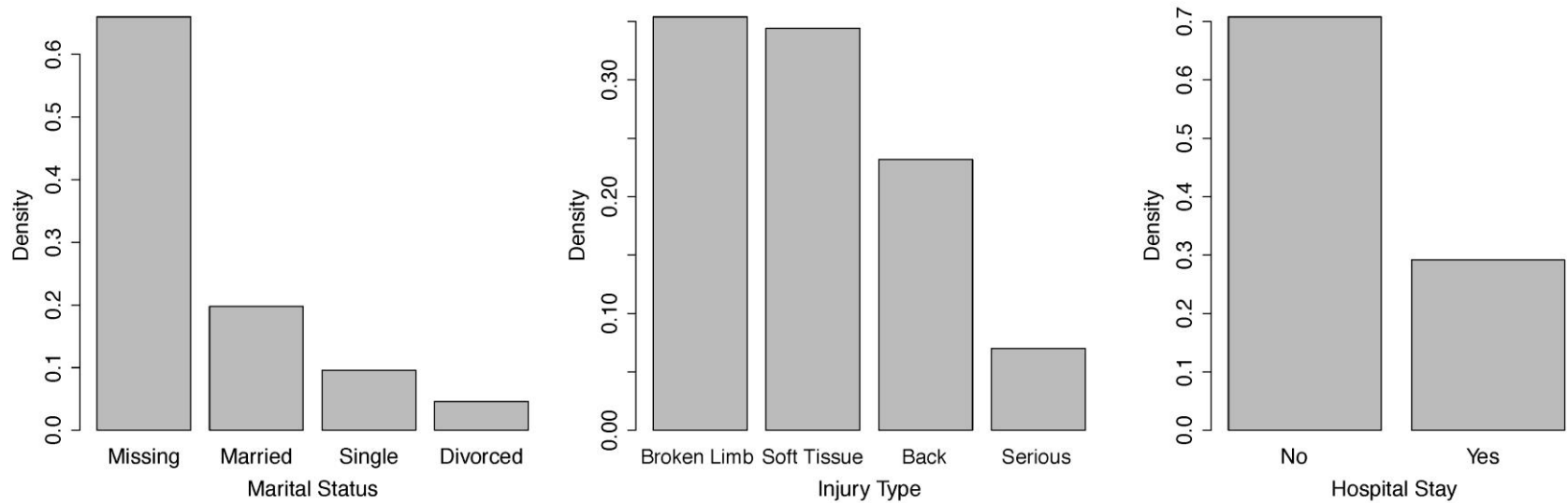


Figure 3.1(k~m): Visualizations of the continuous and categorical features in the motor insurance claims fraud detection ABT in Table 2

CONTENTS

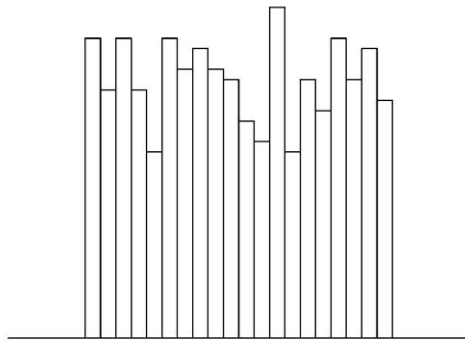
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- For **categorical features**, we should:
 - Examine the **mode**, **2nd mode**, **mode %**, and **2nd mode %** , as these tell us the most common levels within these features and will identify if any levels dominate the dataset.
- For **continuous features** we should:
 - Examine the **mean** and **standard deviation** of each feature to get a sense of the **central tendency** and **variation** of the values within the dataset for the feature.
 - Examine the **minimum** and **maximum** values to understand the **range** that is possible for each feature.

- When generating **histograms** of features, there are a number of common, well understood shapes that we should look out for.

uniform distribution:

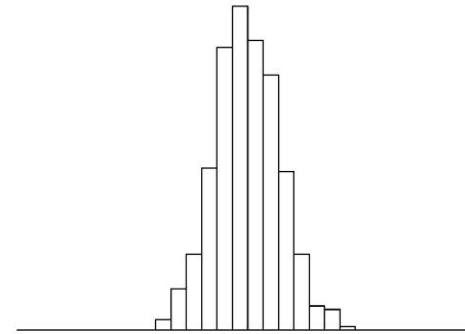
equally likely to take a value in any of the ranges present.



(a) Uniform

normal distribution:

a strong tendency towards a central value and symmetrical variation to either side of this. e. g., the **heights** or **weights** of a randomly selected group of men or women

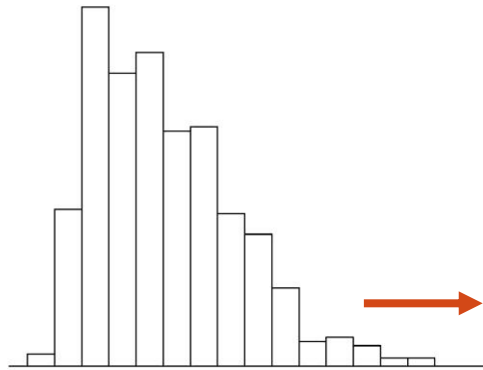


(b) Normal (Unimodal)

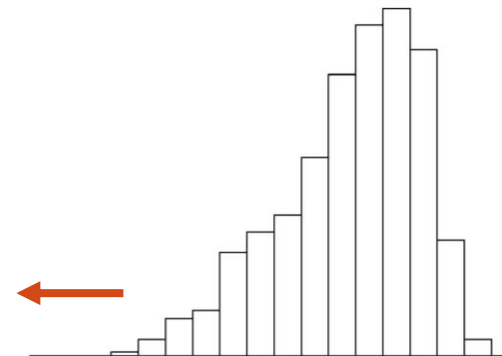
Figure 3.2-1: Histograms for different sets of data each of which exhibit well-known, common characteristics.

Skew:

a tendency towards very high (**right skew**) or very low (**left skew**) values. Skewed distributions have **long tails** toward these very high or very low values.



(c) Unimodal (skewed **right**)

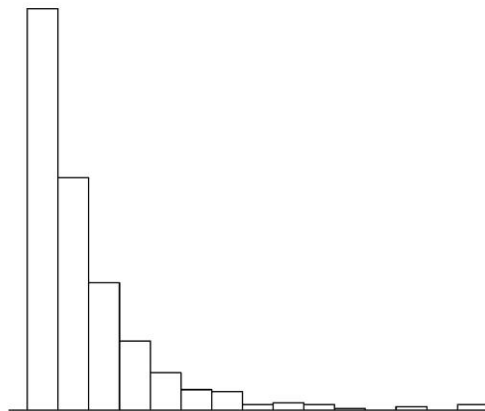


(d) Unimodal (skewed **left**)

Figure 3.2-2: Histograms for different sets of data each of which exhibit well-known, common characteristics.

exponential distribution:

the likelihood of occurrence of a small number of low values is very high, but sharply diminishes as values increase.

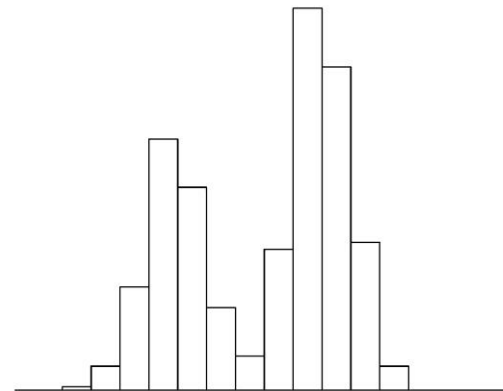


(e) Exponential

multimodal distribution:

has two or more very commonly occurring ranges of values that are clearly separated.

Multimodal distributions tend to occur when a feature contains a measurement made across a number of distinct groups



(f) Multimodal

Figure 3.2-3: Histograms for different sets of data each of which exhibit well-known, common characteristics.

- The probability density function for the **normal** distribution (or **Gaussian distribution**) is

$$N(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where x is any value, and μ and σ are parameters that define the shape of the distribution: the **population mean** and **population standard deviation**.

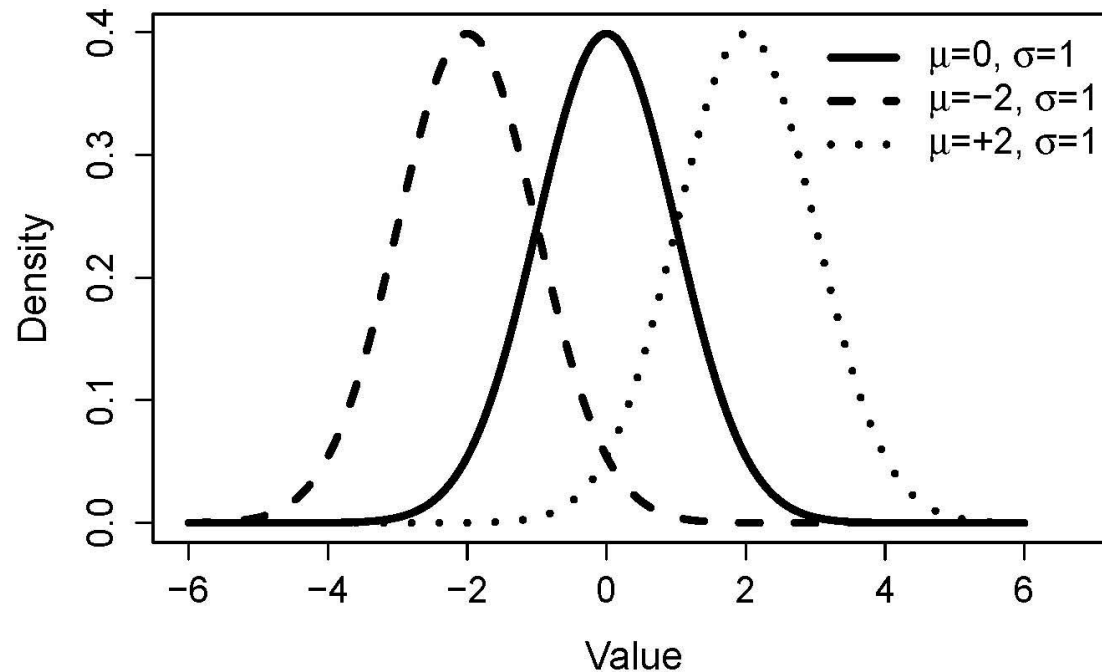


Figure 3.3-1: Three normal distributions with **different means** but **identical standard deviations**.

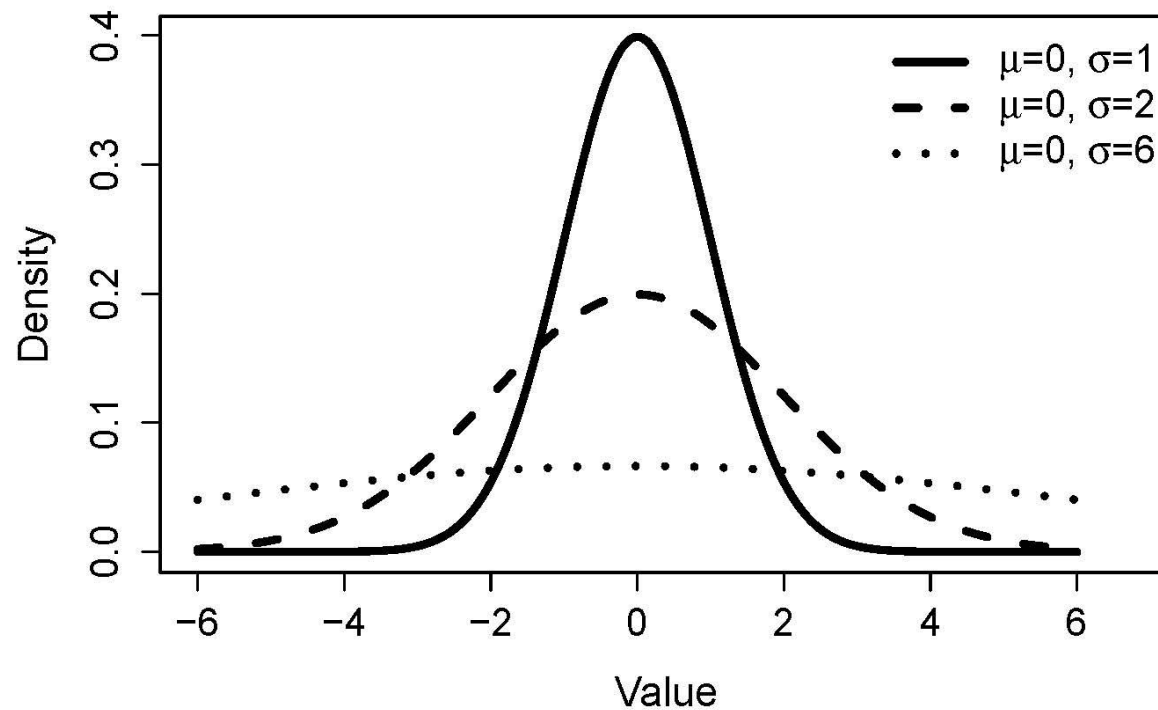
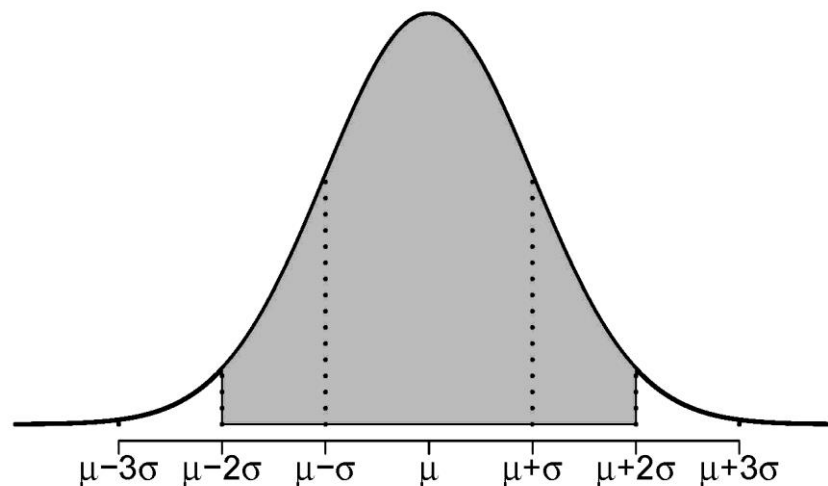


Figure 3.3-2: Three normal distributions with **identical means** but **different standard deviations**.

- The normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$ is known as the **standard normal distribution**.
- The **68 – 95 – 99.7 rule** is a useful characteristic of the normal distribution.
- The rule states that approximately:
 - 68% of the observations will be within **one** σ of μ
 - 95% of observations will be within **two** σ of μ
 - 99.7% of observations will be within **three** σ of μ



CASE STUDY: MOTOR INSURANCE FRAUD

- Examine the [data quality report](#) for the motor insurance fraud prediction scenario and comment on the central tendency and variation of each feature.

- **Table 3.3:** A [data quality report](#) for the motor insurance claims fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

- **Table 3.3:** A [data quality report](#) for the motor insurance claims fraud detection ABT

(a) Continuous

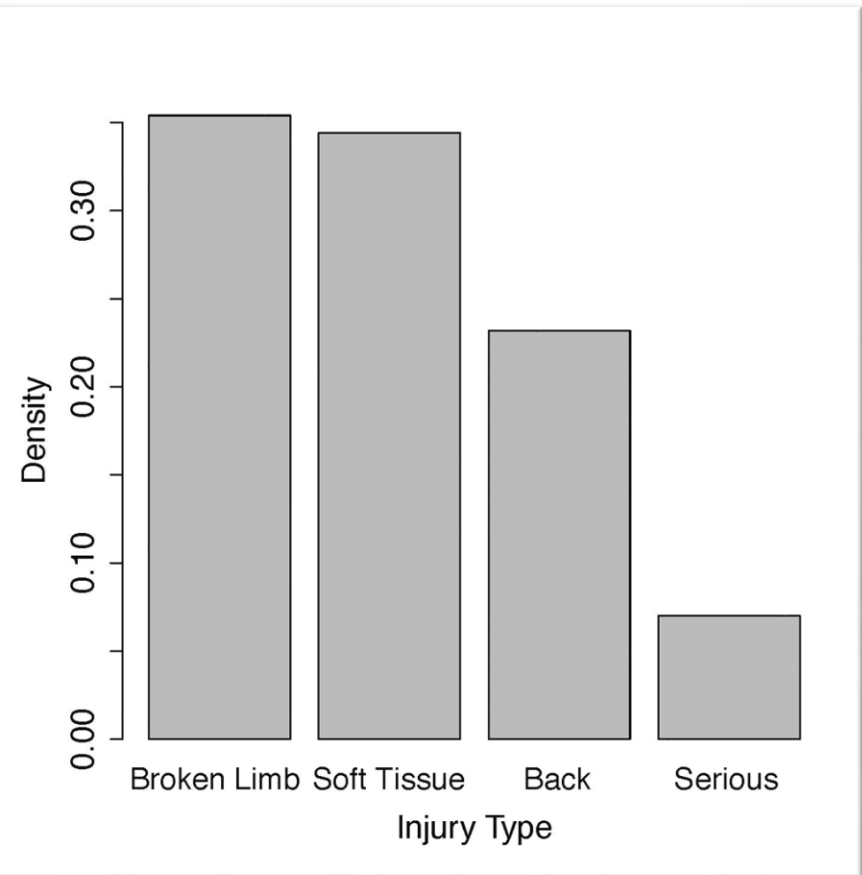
Feature	Count	Miss. %	Card.	Min
INCOME	500	0.0	171	0.0
NUM CLAIMANTS	500	0.0	4	1.0
CLAIM AMOUNT	500	0.0	493	-99,999
TOTAL CLAIMED	500	0.0	235	0.0
NUM CLAIMS	500	0.0	7	0.0

broken limb, soft tissue, and back, are quite frequent, while serious is quite rare

(b) Categorical

Feature	Count	Miss. %	Card.	Mod
INSURANCE TYPE	500	0.0	1	CI
MARITAL STATUS	500	61.2	4	Marri
INJURY TYPE	500	0.0	4	Broken Limb
HOSPITAL STAY	500	0.0	2	No

Broken Limb	177	35.4	Soft Tissue	172	34.4
Back	146	29.2	Serious		



- **Table 3.3:** A [data quality report](#) for the motor insurance claims fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

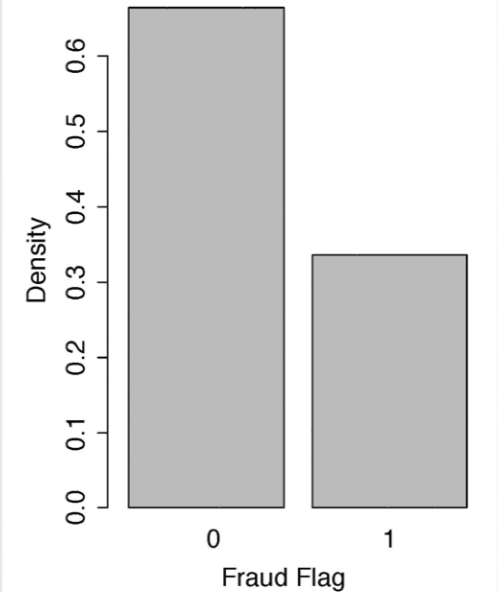
(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

- **Table 3.3:** A **data quality report** for the motor fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0



The distribution of the FRAUD FLAG feature that can be seen in its histogram is **not typical of a continuous feature**

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

- **Table 3.3:** A [data quality report](#) for the motor insurance claims fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

(b) Categorical Features

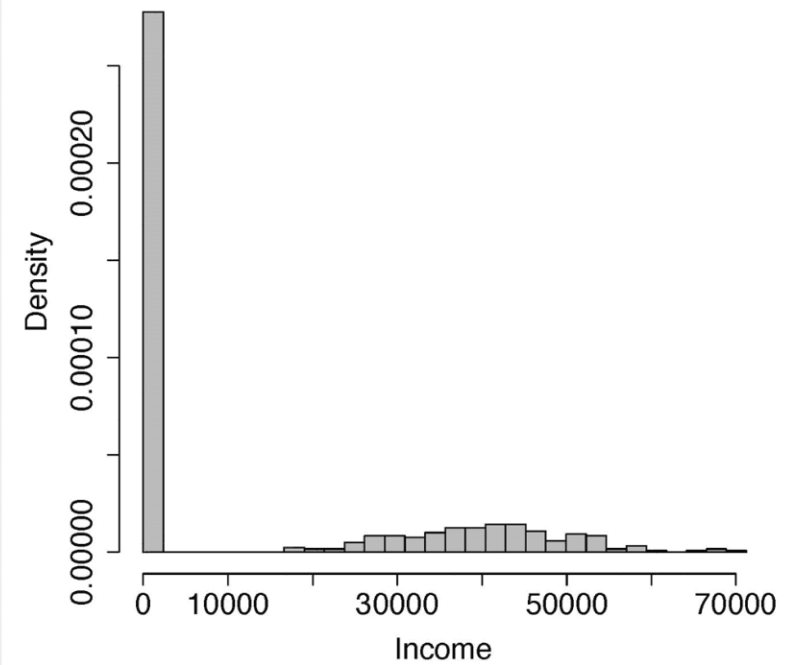
Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

- **Table 3.3:** A **data quality report** for the motor insurance claims fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0					
CLAIM AMOUNT	500	0.0	493	-99,999	3,322					
TOTAL CLAIMED	500	0.0	235	0.0	0.0					
NUM CLAIMS	500	0.0	7	0.0	0.0					
NUM SOFT TISSUE	500	2.0	6	0.0	0.0					
% SOFT TISSUE	500	0.0	9	0.0	0.0					
AMOUNT RECEIVED	500	0.0	329	0.0	0.0					
FRAUD FLAG	500	0.0	2	0.0	0.0					

INCOME is interesting as it seems to follow what looks like a normal distribution except that there is one large bar at about 0.



Feature	Count	Miss.	Card.	Mode
INSURANCE TYPE	500	0.0	1	CI
MARITAL STATUS	500	61.2	4	Married
INJURY TYPE	500	0.0	4	Broken Lim
HOSPITAL STAY	500	0.0	2	No

CONTENTS

- **The Data Quality Report**
- **Getting To Know The Data**
- **Identifying Data Quality Issues**
- **Handling Data Quality Issues**
- **Summary**

- A **data quality issue** is loosely defined as anything **unusual** about the data in an ABT.
- The data quality issues we identify from a data quality report will be of two types:
 - due to **invalid data**.
 - typically arise because of **errors** in the process used to generate an ABT
 - take immediate action to correct them, **regenerate** the ABT, and **recreate** the data quality report.
 - due to **valid data**.

- The most common data quality issues are:
 - **missing values**
 - **irregular cardinality**
 - **Outliers**

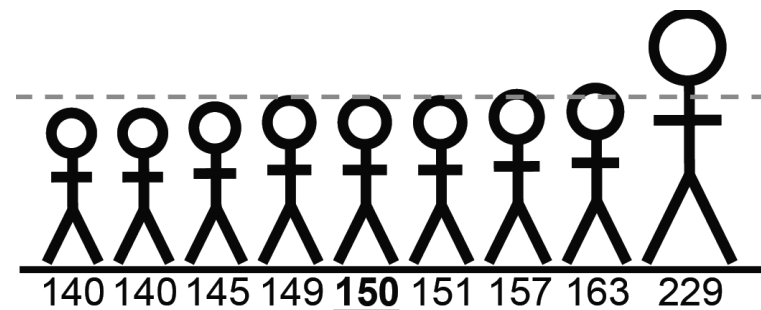
IDENTIFYING IRREGULAR CARDINALITY

- A data quality issue arises when the cardinality for a feature does not match what we expect, a mismatch called **irregular cardinality**.
- Identify:
 - check the cardinality column for are features with a **cardinality of 1**
 - check the cardinality column for categorical features **incorrectly labeled** as continuous
 - check if a categorical feature has a **much higher** cardinality than we would expect given the definition of the feature
 - check when a categorical feature simply has a **very high number** of levels—anything **over 50** is worth investigation
 - ...

IDENTIFYING OUTLIERS

- **Outliers** are values that lie far away from the central tendency of a feature.
- Identify:
 - examine the **minimum** and **maximum** values for each feature and use **domain knowledge** to determine whether these are plausible values.
 - compare the gaps between the **median**, **minimum**, **maximum**, **1st quartile**, and **3rd quartile** values.

*If the gap between the **3rd quartile** and the **maximum** value is **noticeably larger** than the gap between the **median** and the **3rd quartile**, the maximum value is likely to be an outlier.*



CASE STUDY: MOTOR INSURANCE FRAUD

- **Data Quality Issues:**

- Missing Values

- NUM. SOFT TISSUE: **only 2%**
 - MARITAL STATUS: **over 60%**
 - INCOME: **unusual number of zero values**

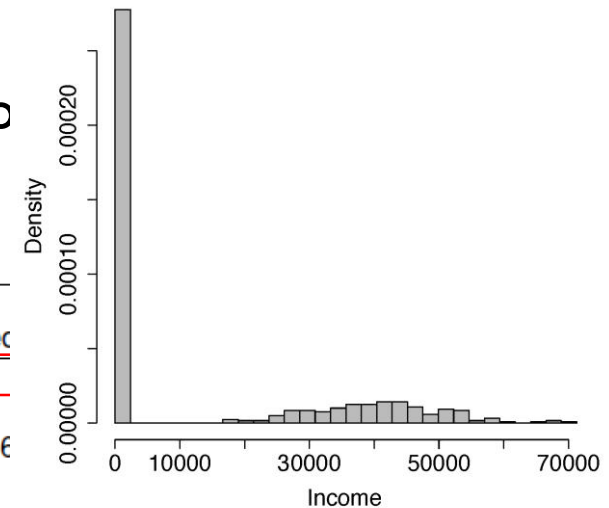
- **Irregular Cardinality**

- **Outliers**

- **Table 3.3:** A data quality report for the moto fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Med
INCOME	500	0.0	171	0.0	0.0	13,740.0	
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,66
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0 0.0 5.0 0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0 0.0 2.0 0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5 8,191.8 295,303.0 30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0 1.0 1.0 0.5



(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

CASE STUDY: MOTOR INSURANCE FRAUD

■ Data Quality Issues:

■ Missing Values

■ Irregular Cardinality

- INSURANCE TYPE: the cardinality is 1, *ci* = car insurance
- NUM. CLAIMANTS/NUM. CLAIMS/NUM. SOFT TISSUE/
% SOFT TISSUE: all have cardinality less than 10
- FRAUD FLAG: the cardinality is 2, target feature, need to be changed to a categorical feature.

■ Outliers

- **Table 3.3:** A **data quality report** for the motor insurance claims fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

CASE STUDY: MOTOR INSURANCE FRAUD

■ Data Quality Issues:

■ Missing Values

■ Irregular Cardinality

■ Outliers

- CLAIM AMOUNT: (ID3) an unusual minimum value of **-99,999, error!**
- TOTAL CLAIMED/NUM. CLAIMS : (ID460) unusually high maximum values, a **company policy** rather than an individual policy
- CLAIM AMOUNT / AMOUNT RECEIVED: (ID302) unusually high maximum values, **serious injury**.
- DQI→ABT

- **Table 3.3:** A [data quality report](#) for the motor insurance claims fraud detection ABT

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2 nd Mode	2 nd Mode Freq.	2 nd Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

- **Table 3.2:** Portions of the **ABT** for the motor insurance claims fraud detection problem.

ID	TYPE	INC.	MARITAL STATUS	NUM CLMNTS.	INJURY TYPE	HOSPITAL STAY	CLAIM AMNT.	TOTAL CLAIMED	NUM CLAIMS	NUM SOFT TISS.	% SOFT TISS.	CLAIM AMT RCVD.	FRAUD FLAG
1	CI	0	Married	2	Soft Tissue	No	1,625	3250	2	2	1.0	0	1
2	CI	0		2	Back	Yes	15,028	60,112	1		0	15,028	0
3	CI	54,613		1	Broken Limb	No	-99,999	0	0	0	0	572	0
4	CI	0		4	Broken Limb	Yes	5,097	11,661	1	1	1.0	7,864	0
5	CI	0	Single	4	Soft Tissue	No	8869	0	0	0	0	0	1
6	CI	0		1	Broken Limb	Yes	17,480	0	0	0	0	17,480	0
7	CI	52,567		3	Broken Limb	No	3,017	18,102	2	1	0.5	0	1
8	CI	0		2	Back	Yes	7463	0	0	0	0	7,463	0
9	CI	0	Married	1	Soft Tissue	No	2,067	0	0	0	0	2,067	0
10	CI	42,300		4	Back	No	2,260	0	0	0	0	2,260	0
⋮													
300	CI	0	Married	2	Broken Limb	No	2,244	0	0	0	0	2,244	0
301	CI	0		1	Broken Limb	No	1,627	92,283	3	0	0	1,627	0
302	CI	0		3	Serious	Yes	270,200	0	0	0	0	270,200	0
303	CI	0		1	Soft Tissue	No	7,668	92,806	3	0	0	7,668	0
304	CI	46,365	1	Back	No	3,217	0	0	0	0	1,653	0	
⋮													
458	CI	48,176	Married	3	Soft Tissue	Yes	4,653	8,203	1	0	0	4,653	0
459	CI	0	Divorced	1	Soft Tissue	Yes	881	51,245	3	0	0	0	1
460	CI	0		3	Back	No	8,688	729,792	56	5	0.08	8,688	0
461	CI	47,371		1	Broken Limb	Yes	3,194	11,668	1	0	0	3,194	0
462	CI	0		1	Soft Tissue	No	6,821	0	0	0	0	0	1
⋮													
491	CI	40,204	Single	1	Back	No	75,748	11,116	1	0	0	0	1
492	CI	0	Married	1	Broken Limb	No	6,172	6,041	1		0	6,172	0
493	CI	0		1	Soft Tissue	Yes	2,569	20,055	1	0	0	2,569	0
494	CI	31,951		1	Broken Limb	No	5,227	22,095	1	0	0	5,227	0
495	CI	0		2	Back	No	3,813	9,882	3	0	0	0	1
496	CI	0	Married	1	Soft Tissue	No	2,118	0	0	0	0	0	1
497	CI	29,280		4	Broken Limb	Yes	3,199	0	0	0	0	0	1
498	CI	0		1	Broken Limb	Yes	32,469	0	0	0	0	16,763	0
499	CI	46,683		1	Broken Limb	No	179,448	0	0	0	0	179,448	0
500	CI	0		1	Broken Limb	No	8,259	0	0	0	0	0	1

Table 3.4: The structure of a data quality plan.

Feature	Data Quality Issue	Potential Handling Strategies
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

CASE STUDY: MOTOR INSURANCE FRAUD

- **Table 3.5:** The data quality plan for the motor insurance fraud prediction ABT.

Feature	Data Quality Issue	Potential Handling Strategies
NUM SOFT TISSUE	Missing values (2%)	
CLAIM AMOUNT	Outliers (high)	
AMOUNT RECEIVED	Outliers (high)	

CONTENTS

- **The Data Quality Report**
- **Getting To Know The Data**
- **Identifying Data Quality Issues**
- **Handling Data Quality Issues**
- **Summary**

HANDLING MISSING VALUES

- Approach 1: Drop any features that have missing value.
- Approach 2: Apply **complete case analysis**.
- Approach 3: Derive a **missing indicator feature** from features with missing value.

HANDLING MISSING VALUES

- **Imputation** replaces missing feature values with a **plausible estimated value** based on the feature values that are present.
- The most common approach to imputation is to replace missing values for a feature with **a measure of the central tendency** of that feature.
- We would be **reluctant** to use imputation on features missing in excess of **30%** of their values and would **strongly recommend against** the use of imputation on features missing in excess of **50%** of their values.

HANDLING OUTLIERS

- The easiest way to handle outliers is to use a **clamp transformation** that clamps all values above an **upper threshold** and below a **lower threshold** to these threshold values, thus removing the offending outliers

$$a_i = \begin{cases} lower & \text{if } a_i < lower \\ upper & \text{if } a_i > upper \\ a_i & \text{otherwise} \end{cases}$$

where a_i is a specific value of feature a ,
lower and upper are the lower and upper thresholds.

HANDLING OUTLIERS

- One common way: set the **lower threshold** to the 1st quartile value minus 1.5 times the **inter-quartile range** and the **upper threshold** to the 3rd quartile plus 1.5 times the inter-quartile range.
- Be used for the **CLAIM AMOUNT** feature from the motor claims insurance fraud detection scenario, then the upper and lower thresholds would be:

$$\text{lower} = 3,322.3 - 1.5 \times 8,923.2 = -10,062.5$$

$$\text{upper} = 12,245.5 + 1.5 \times 8,923.2 = 25,630.3$$

Feature	Count	% Miss.	Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3

HANDLING OUTLIERS

- Another commonly used approach: use the **mean value** of a feature plus or minus **2 times the standard deviation**.
- Be used for the **AMOUNT RECEIVED** feature from the motor claims insurance fraud detection scenario, the upper and lower thresholds would be :

$$\text{lower} = 13,051.9 - 2 \times 30,547.2 = -48,042.5$$

$$\text{upper} = 13,051.9 + 2 \times 30,547.2 = 74,146.3$$

Feature	Count	% Miss.	% Card.	Min	1 st Qrt.	Mean	Median	3 rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

CASE STUDY: MOTOR INSURANCE FRAUD

- **Table 3.5:** The data quality plan for the motor insurance fraud prediction ABT.

Feature	Data Quality Issue	Potential Handling Strategies
NUM SOFT TISSUE	Missing values (2%)	Imputation (median: 0.0)
CLAIM AMOUNT	Outliers (high)	Clamp transformation (manual: 0, 80 000)
AMOUNT RECEIVED	Outliers (high)	Clamp transformation (manual: 0, 80 000)

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- The key outcomes of the **data exploration** process are that the practitioner should
 - Have gotten to know the features within the ABT, especially their **central tendencies**, **variations**, and **distributions**.
 - Have identified any **data quality issues** within the ABT, in particular **missing values**, **irregular cardinality**, and **outliers**.
 - Have corrected any data quality issues due to **invalid data**.
 - Have recorded any data quality issues due to **valid data** in a **data quality plan** along with potential handling strategies.
 - Be confident that enough good quality data exists to continue with a project.

HOMEWORK

- Reading: Section 3.1~3.4

Appendix A